More SQL

MBUA 512

SQL = Structured Query Language, "es-queue-el" or "sequel". With Michael Winikoff.

- SELECT calculations - and DISTINCT
- WHERE conditions
- ORDER BY (DESC)
- GROUP BY, and aggregates (min, max, avg, count, sum)
- LIMIT n

How to write SQL queries & tips

Other things(?): Transactions?

Based on slides by Professor Markus Luczak-Roesch.

Recap: SQL commands

- `CREATE TABLE` and `ALTER TABLE` – define the structure of a table (its *schema*)
- `INSERT INTO`, `DELETE FROM`, `UPDATE` – change the data in the table
- `SELECT` – query data

(there are a lot more than these)

> Important: make sure you select `MySQL` (not `Trino`) and your own workspace (`user_[username]`)

SELECT calculations

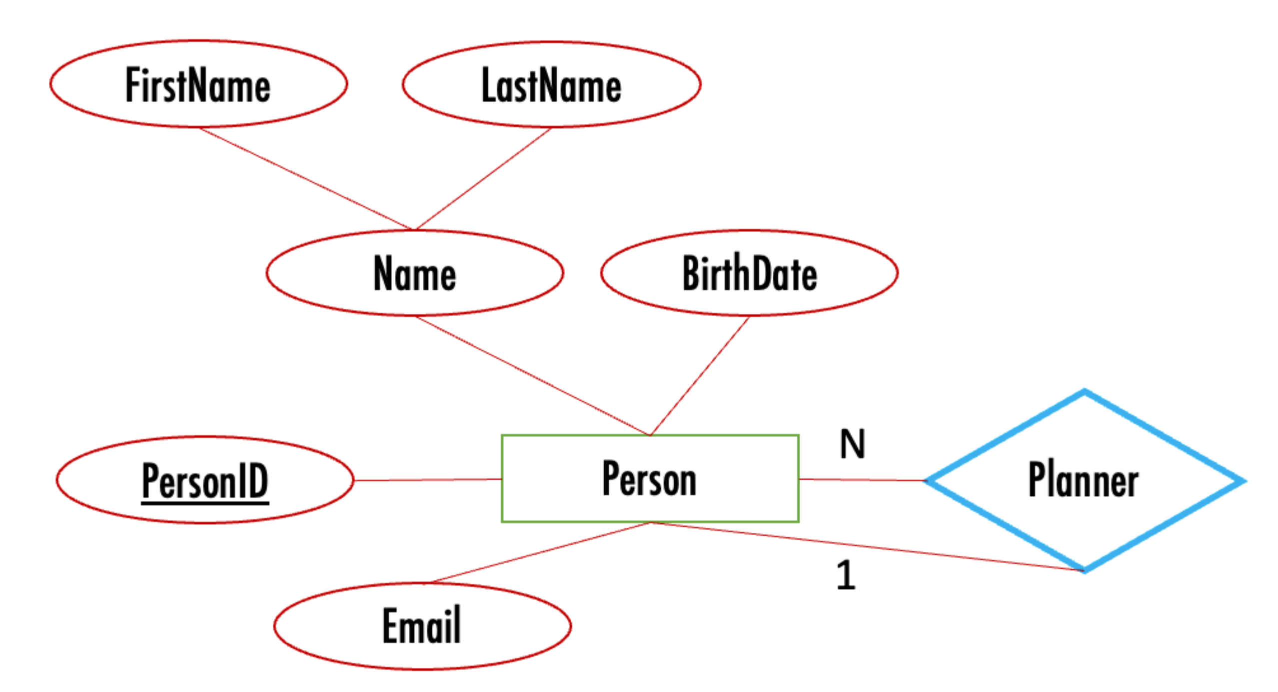
- string `||`
- AS
- numerical operations including floor, ceiling, round
- date operations - including current date, subtraction

WHERE conditions

- IS NULL
- IN
- LIKE and `%_`
- numerical conditions
- date conditions
- logical operators (AND, OR, NOT)
- (not) EXISTS
- UNION, INTERSECT

Running Example: birthday

- Scenario: send a message to people on their birthday, and remind their designated party planner to plan a party earlier
- Data: name (first, last), birthdate, email address, and designated party planner



To create the database

```sql
DROP TABLE IF EXISTS  people;

CREATE TABLE people (
         personID INTEGER PRIMARY KEY AUTOINCREMENT,
         firstName VARCHAR(30) NOT NULL,
         lastName VARCHAR(30),
         birthdate DATE NOT NULL,
         planner INTEGER,
         email VARCHAR(30),
         FOREIGN KEY(planner) REFERENCES people(personID)
);
```

To add data

```
INSERT INTO people VALUES
          (1, 'Bob', 'Smith', '1917-01-03', 1, NULL),
          (2, 'Bob', 'Smith', '1973-03-25', NULL, 'bob@smith.net'),
          (3, 'Alice', 'Smith', '2026-07-07', 1, 'Alice@smith.net'),
          (4, 'sam', 'Smith', '2020-01-03', 1, 'sam'),
          (5, 'Bobbie','Smithton', '1987-03-16', NULL, 'bob@smith.net');
```

Some queries to write (1/2)

1. Find people whose birthday is today so we can email them (combine first and last names)
2. Find people with a special birthday (e.g. 100) coming up (hint: need to calculate age, use FLOOR)
3. Find people with a birthday in January, February or March so we can email their planner (hint: use IN)

Some queries to write (2/2)

administrative tasks:

4. Find people with missing or invalid email addresses (hint: use NULL AND ... NOT LIKE)
5. Find where two (or more) people have the same email address (hint: use EXISTS)
6. Find people who are not planners for anyone else and do not have a birthday (hint: use EXISTS)

analysis task:

7. how many people have a birthday in each month? (hint: use GROUP BY and EXTRACT(MONTH from ...))

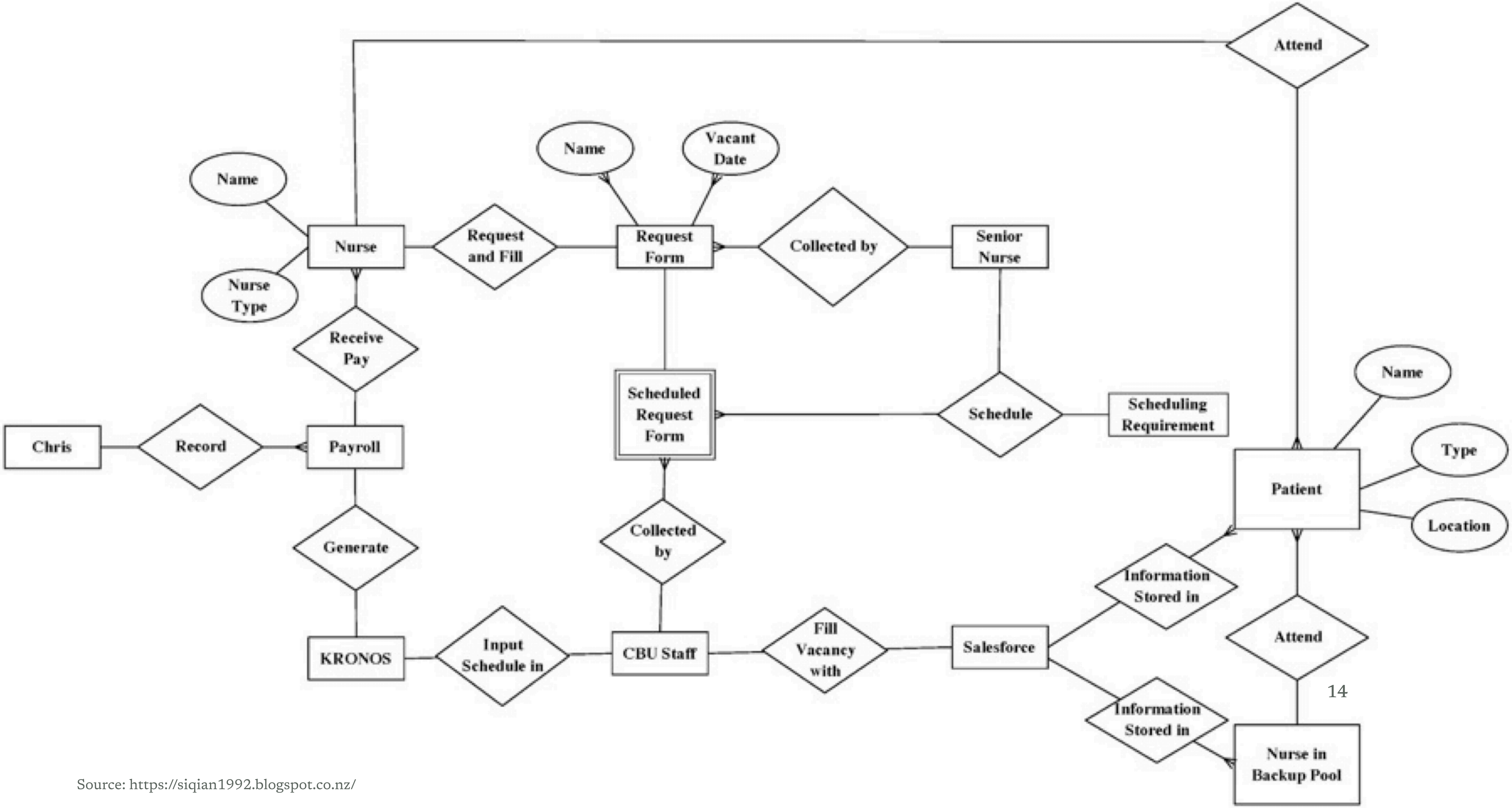
Tips for writing SQL

- Build it up in pieces, testing each piece (`*` can be useful)
- Think about:
  - What information you need to see in the result (which columns ...) ... values? Calculations? Aggregate calculations (e.g. SUM)?
  - What tables they come from
  - How you need to JOIN the tables
  - What constraints to use ... and how to express them
    - WHERE if based on row
    - WHERE ... AND exists – if not

More tips

- `' '` (single quotes), not `"` (double quote)
- Distinct – the whole row needs to be distinct, so sometimes select fewer columns
- Group By … Having … - the ”Having” goes just after the “Group By …”
- NULL is not the same as the empty string
- Too many rows in result? Check JOIN conditions (or using “NATURAL JOIN” - which is not the same as “JOIN”)
- No rows in result? Check conditions, and perhaps remove some
- Using exists – make sure there is also an “AS …” (so can refer to parent query in sub-query)

Summary



Source: https://siqian1992.blogspot.co.nz/

Some useful Superset commands (MOVE OR DELETE)

```
show schemas from idrive
show tables in idrive.relbench_rel_ratebeer   -- most tables
show tables in idrive.relbench_rel_avito
select * from idrive.relbench_rel_ratebeer.countries
```

Select schema `user_[username]`, replacing `[username]` with your username (e.g. for me, `user_michaelwinikoff`).

Activities

Some possible activities for class 3

- CREATE a DB
- DB content:
    - Add content - including from CSV?
    - Delete content
    - Update content
- Explore constraints
- Simple (one table) SELECT
- Joins

Some possible activities for class 4

- to complete

Activity – ...

~ 10 minutes, in groups